$$R'_A = R_A + K(S_A - E_A)$$

K is a constant (max adj. rating per game)

Advantages of ELO sys.	Disadvantages
- pts. always balanced (sum the same)	- doesn't take into account individual contributions of team members.
- self-corrected sys. weaker player wins, ↑ pts. gained	- Rating deflation - good rating requires skill over time to maintain.
- works w/ team games average for team used as elo	- Can't compare rating history

Unit 8 PPO - Proximal Policy Optimization

↳ improves training stability by avoiding too large of policy changes.

- Smaller policy updates are more likely to converge at the optimal solution.
- Too big of an update can result in a bad policy w/ no possibility of recovery.

PPO updates the policy conservatively clip new policy to old policy in range

$$[1-\varepsilon, 1+\varepsilon]$$

remove the policy incentive for too large of changes.

PPO constrains policy updates w/ new objective func. Clipped surrogate objective function

Clipped surrogate objective function is designed to avoid destructively large weight updates.

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon\right.$$

ratio function: $r_t(\theta) = \dfrac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)}$

the probability of doing a_t in s_t for the current policy divided by the previous policy - estimate the chance of divergence b/w old & new policy

2 solutions to clip policy:

- TRPO (Trust Region Policy Optimization) uses KL divergence constraints outside the objective function to constrain the policy update.
- PPO clip probability ratio directly in the objective function

taking min of clipped & act. change for a pessimistic bound.